

P.E.T.R.A: Persuasive Environment for Tracking and Regulating Arousal and Valence in Virtual Reality

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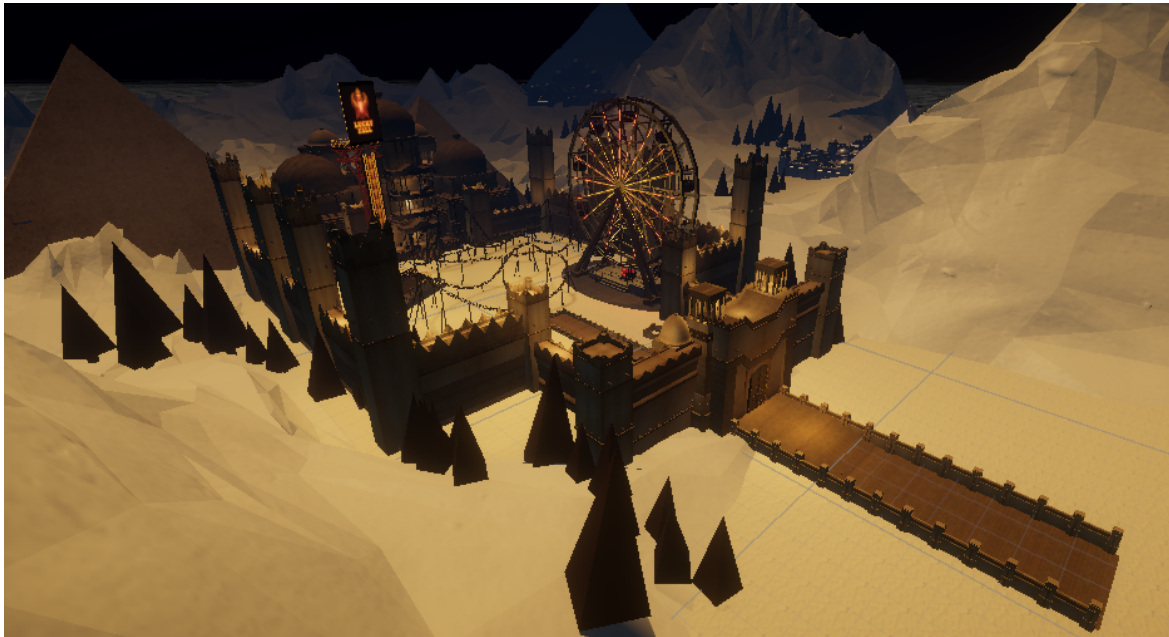


Figure 1: The virtual environment of P.E.T.R.A., featuring attractions designed to modulate emotional arousal and valence.

ABSTRACT

The rising prevalence of gambling, particularly in digital formats, demands innovative approaches to understanding and mitigating Gambling Disorder (GD). Virtual Reality (VR), with its immersive and interactive potential, offers a unique platform to explore the complex interplay of psychological and environmental factors contributing to impulsive gambling behaviors. This research introduces **P.E.T.R.A. (Persuasive Environment for Tracking and Regulating Arousal)**, a novel VR-based environment designed as an amusement park simulation. P.E.T.R.A. combines elements inspired by electronic gambling machines (EGMs) with gamification mechanics to study and modulate emotional arousal and valence. Using the Circumplex Model of Emotion, P.E.T.R.A. tracks and analyzes affective states in real-time, providing insights into impulsive decision-making. Preliminary results validate P.E.T.R.A.'s capability to evoke realistic emotional responses and inform future interventions, particularly for cognitive behavioral therapy (CBT). This work aims at demonstrating the potential of VR technologies for GD research, prevention, and therapeutic applications.

Index Terms: Virtual Reality, Gambling Disorder, Emotional Regulation, Gamification, Cognitive Behavioral Therapy.

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1 INTRODUCTION

Gambling Disorder (GD) is a behavioral addiction classified by the DSM-5, the Diagnostic and Statistical Manual of Mental Disorders (1), as a significant psychological condition, with symptoms including persistent gambling despite adverse consequences, loss of control, and preoccupation with gambling activities. The rapid growth of online gambling platforms and their accessibility through mobile devices have exacerbated the problem, making it a global public health concern. Vulnerable groups, such as young adults, individuals with impulsive tendencies, and those with a history of mental health conditions, are disproportionately affected (2).

This paper introduces **P.E.T.R.A.**, a VR-based environment designed to simulate a variety of gambling scenarios in a controlled, immersive amusement park setting. By leveraging the Circumplex Model of Emotion (4), P.E.T.R.A. provides a novel method for tracking and regulating emotional responses, offering valuable insights into the psychological mechanisms underlying GD.

2 SYSTEM ARCHITECTURE

P.E.T.R.A. integrates advanced VR technologies, real-time emotional tracking, and a modular design to study the interplay between emotional arousal, valence, and decision-making behaviors in gambling-like scenarios. By adjusting parameters such as ride speed, sensory feedback, and reward probabilities, the platform explores how different environmental stimuli influence emotional states and impulsive behaviors.

The system is designed to address key research questions, including how emotional arousal and valence vary during interactions

with immersive attractions, and how these emotional trajectories correlate with impulsive decision-making. Additionally, P.E.T.R.A. investigates the potential of combining VR with real-time monitoring technologies, like EEG and GSR, to improve the precision and adaptability of therapeutic interventions.

P.E.T.R.A.'s modularity allows for personalized configurations, tailoring the experience to individual users' responses. This flexibility is particularly valuable for developing customized cognitive-behavioral therapies (CBT) that adapt based on users' emotional reactions, with guidance from psychological experts. The platform's ability to provide dynamic, data-driven interventions makes it a powerful tool for both research and practical therapeutic applications.

2.1 Emotion-Driven Attractions

P.E.T.R.A.'s virtual amusement park is designed to regulate emotional arousal and valence through six carefully crafted experiences. Each attraction serves a specific purpose in understanding and influencing emotional and behavioral responses.

The **Roller Coaster (High Stakes)** and **Drop Tower (Lucky Fall)** evoke heightened arousal through immersive audio-visual effects and dynamic movements, simulating the excitement and tension of gambling. In contrast, the **Ferris Wheel (Wagering Wheel)** provides a calming experience, fostering introspection and emotional regulation.

For valence modulation, mini-games like the **Plinko Board (Bouncing Odds)** and **Spinning Wheel (Illusion Wheel)** explore reward anticipation and disappointment, while the **Dice Game (Dicey Choices)** emphasizes the illusion of control. Together, these attractions form a dynamic environment for studying emotional regulation and impulsive decision-making, with potential therapeutic applications.

2.2 Real-Time Emotional Tracking

A key innovation of P.E.T.R.A. is its integration of the Circumplex Model of Emotion (4), which conceptualizes emotions along two dimensions: valence (pleasantness) and arousal (intensity). Users actively interact with a dynamic interface to pinpoint their current emotional state from 13 fundamental emotions.

The system records these interactions in two formats:

- **Tabular Data:** Emotional states are logged in structured tables, enabling longitudinal analysis of patterns across multiple sessions.
- **Visual Snapshots:** Screenshots of the Circumplex Model capture real-time emotional dynamics, offering intuitive visual feedback for both researchers and participants.

Future developments include integrating physiological tracking devices such as EEG headsets and GSR bracelets to enhance the precision of emotional monitoring and provide multimodal data for analysis.

3 USABILITY TESTING

The usability of P.E.T.R.A. was evaluated using the System Usability Scale (SUS) (3), a widely accepted tool for assessing user experience. The evaluation involved a diverse group of participants who provided detailed feedback on various aspects of the platform. The SUS results yielded an average score of 77.8, placing P.E.T.R.A. in the "excellent" range for usability. This high score reflects the platform's ability to deliver a seamless and engaging user experience.

Participants provided feedback on:

- **Ease of Use:** The intuitive interface facilitated seamless interaction within the VR environment, with users highlighting the straightforward navigation and clear prompts.

- **Engagement:** High levels of immersion and emotional involvement were reported during gameplay, with several participants noting the realistic design of the attractions as a standout feature.

- **Reliability of Emotional Tracking:** Consistency between user-reported emotions and system-recorded data was verified, further emphasizing the accuracy and reliability of the tracking system.

These results confirm the platform's readiness for broader applications in research and therapy, showcasing its ability to combine usability with advanced emotional tracking capabilities.

4 APPLICATIONS AND FUTURE PROSPECTS

P.E.T.R.A. demonstrates significant potential for advancing research and therapy in emotional regulation and decision-making. Future developments aim to:

- **Advanced Tracking Devices:** Integration of EEG headsets and GSR bracelets to provide multimodal emotional tracking, enhancing the precision of interventions.
- **Machine Learning Integration:** AI algorithms will analyze emotional and behavioral data in real-time, enabling adaptive feedback and personalized therapeutic strategies.
- **Therapeutic Expansions:** Development of specialized modules targeting conditions such as anxiety, depression, and phobias.
- **Collaborative Environments:** Multi-user VR settings will explore social dynamics and peer influences on emotional and behavioral responses.
- **Commercial Deployment:** Partnerships with healthcare providers and educational institutions will scale P.E.T.R.A. for broader use in clinical and preventive applications.

The platform's combination of immersive VR technology, robust emotional tracking, and adaptability positions it as a groundbreaking tool for research and applied psychology, fostering innovative approaches to behavioral health.

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